

MUHAMMAD ADEEL

AI/ML Engineer | Generative AI | LLMs | RAG | AI Agents

Pakistan | +92 309 5408984 | muhammadadeel0a1@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI/ML engineer building systems across machine learning, deep learning, and generative AI — from transfer-learning computer vision models to retrieval-augmented LLM applications and tool-calling AI agents. Experienced across the ML lifecycle, including data preprocessing, model training and evaluation, and API-based deployment. Works fluently in Python, PyTorch/TensorFlow, and LangChain/LangGraph.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MySQL, Pandas, NumPy

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Regression, Classification, Ensemble Learning, Feature Engineering, Cross-Validation

Deep Learning & NLP: PyTorch, TensorFlow, Transformers, NLP, CNN, RNN, Transfer Learning

Generative AI & LLMs: LLMs, RAG, Gemini API, Embeddings, Vector Search, Reranking, Semantic Search, Prompt Engineering, LangChain, LangGraph, AI Agents, Tool Calling

Computer Vision: EfficientNetV2, DenseNet121, YOLO, BLIP, OCR, Image Classification, Object Detection, Image Captioning

Backend & Deployment: FastAPI, Docker, Git, GitHub

PROFESSIONAL EXPERIENCE

AI Engineer — QMH Technologies, Rawalpindi

Sep 2025 – Aug 2026

- Worked full-time, onsite as an AI Engineer, building solutions around LLMs and generative AI frameworks including LangChain and the OpenAI API
- Built and integrated RAG pipelines — document ingestion, chunking, embedding generation, and vector retrieval — to ground LLM outputs in company data
- Developed LLM-based AI agents with tool-calling capabilities to automate multi-step workflows instead of relying on scripted logic
- Applied strong Python fundamentals to translate hands-on GenAI experimentation into production-oriented engineering work within the team

AI Engineer Intern — QMH Technologies, Rawalpindi

Jul 2025 – Sep 2025

- Trained in core AI/ML concepts, assisting in developing and testing models using Python, NumPy, and Pandas for data preprocessing and evaluation
- Explored generative AI fundamentals, including LLMs and prompt engineering, alongside hands-on ML/DL model experimentation

PROJECTS (Personal)

Enterprise AI Knowledge Assistant — RAG / LLM [\[GitHub\]](#)

- Engineered a production-style RAG pipeline — ingestion, chunking, embedding, retrieval and reranking — to ground LLM responses in enterprise documents
- Integrated the Gemini API as the generation layer and deployed the pipeline as a Dockerized FastAPI service with an evaluation process for retrieval and response quality

AI Customer Support Agent — LLM / AI Agents [\[GitHub\]](#)

- Designed a LangGraph-orchestrated agent handling support workflows — intent understanding, knowledge retrieval, and order actions — via multi-step tool calling
- Automated end-to-end resolution of repetitive support queries, from natural language request to completed action, reducing dependency on manual responses

AI-Powered “Third Eye” for Visually Impaired Persons — Multimodal AI [\[GitHub\]](#)

- Built a real-time multi-module assistive system combining DenseNet121, YOLO11n, BLIP captioning, OCR, and text-to-speech, achieving 90% classification accuracy on a Pakistan-specific dataset

Spine Disease Detection System — EfficientNetV2 [\[GitHub\]](#)

- Fine-tuned EfficientNetV2 with transfer learning, data augmentation, and sampling techniques to classify spinal disorders from the RSNA 2024 dataset, achieving 95% accuracy

Heart Disease Prediction — Ensemble ML [\[GitHub\]](#)

- Built and compared classification models (Logistic Regression, Random Forest, SVM, XGBoost, LightGBM, CatBoost) on the Kaggle Playground Series S6E2 dataset, reaching an ensemble AUC of ~0.956

EDUCATION

Bachelor of Computer Science — University of Haripur

Sep 2021 – Jun 2025

CERTIFICATIONS

- Machine Learning Specialization — Andrew Ng (Coursera) [\[Verify\]](#)
- Deep Learning Specialization — Andrew Ng (Coursera) [\[Verify\]](#)